

Learning Food Affordability Representations to Predict Obesity, Diabetes and Healthcare Costs in the United States



Data-driven framework that integrates machine-learned representations, interpretable models, and causal analysis to understand the impact of food affordability on health outcomes and healthcare costs in the United States.

Introduction

- Food affordability is a key determinant of dietary behavior and long-term health outcomes in the United States. Traditional measures of affordability often rely on simplified, linear assumptions that fail to capture complex interactions between income, access, and mobility.
- This project develops a data-driven framework to better understand how food affordability influences obesity, diabetes, and healthcare costs.

Objectives

- Develop data-driven representations of food affordability using machine learning.
- Compare learned representations (Autoencoders, PCA) with traditional variables.
- Evaluate predictive performance and model interpretability.
- Estimate causal effects of affordability on health outcomes.
- Assess potential policy impacts through simulation.

Data Overview

Unit of analysis: U.S. counties (~3,100)

Datasets:

Economic & Socioeconomic

- Median household income
- Poverty rate
- Minimum wage

Health Outcomes

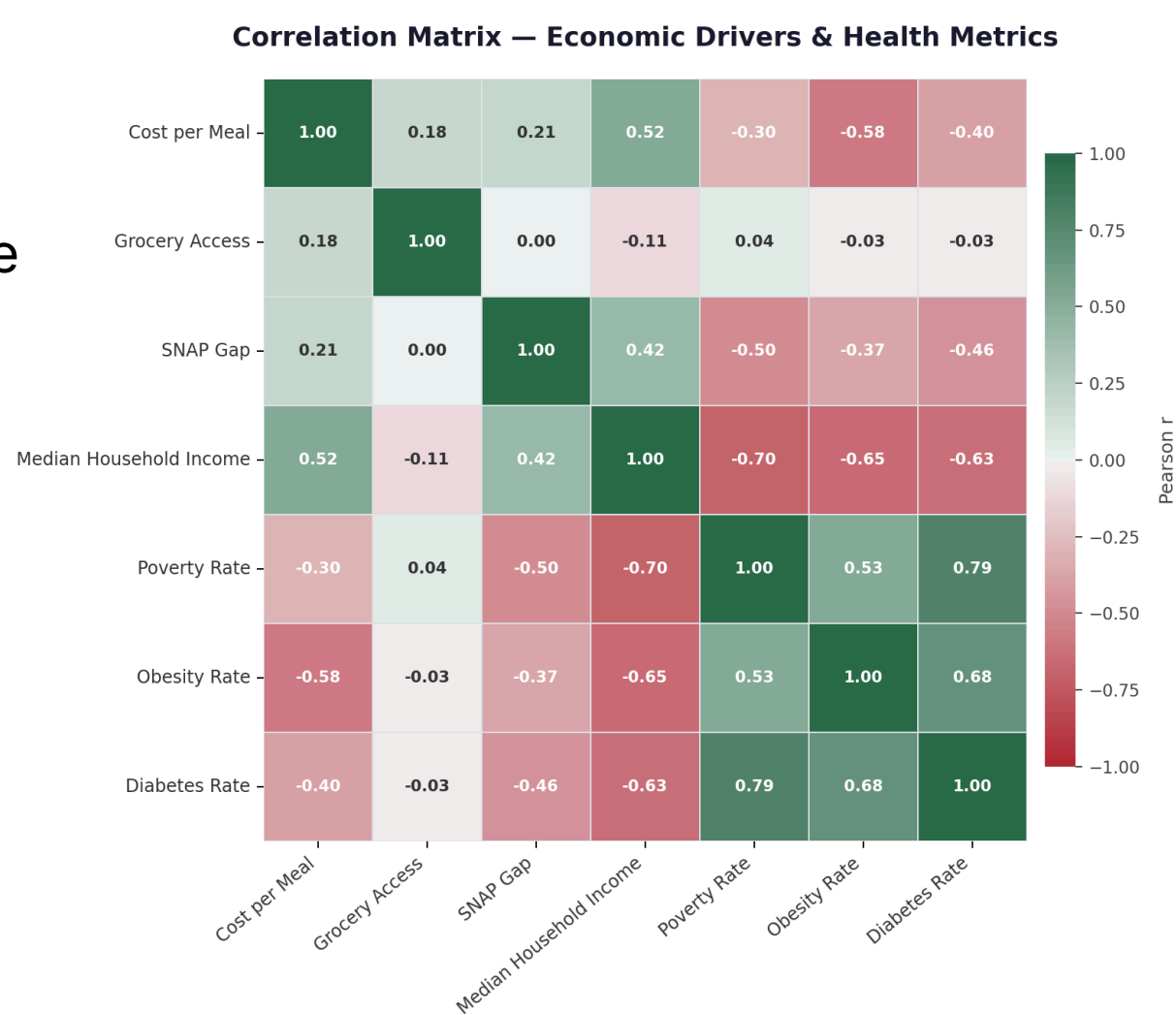
- Obesity rate
- Diabetes rate
- Medicare cost per capita

Food Environment

- Cost per meal
- Grocery store access (GROCPH20)
- SNAP participation & SNAP gap
- % households without vehicle
- Mean commute time
- Spatial Lag Features

Demographics

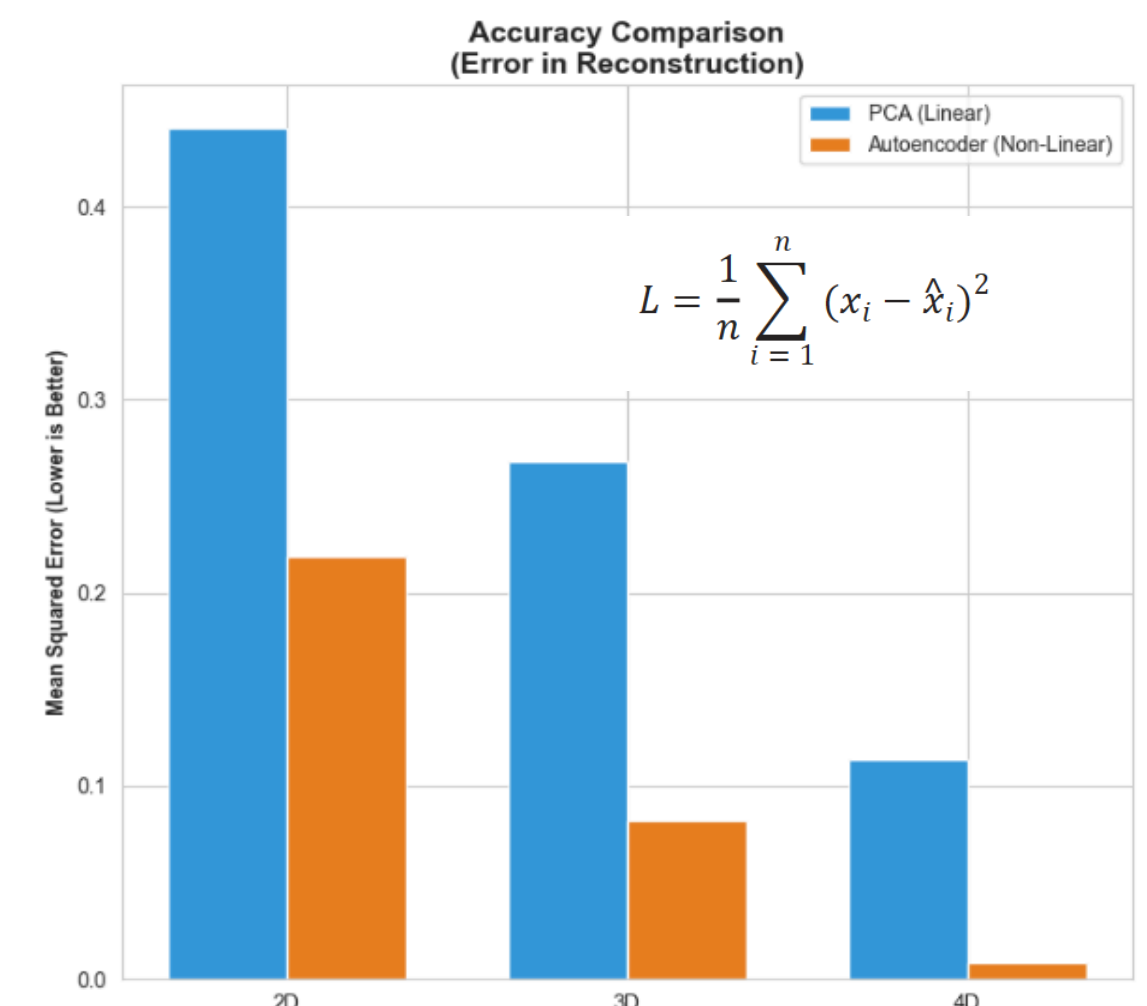
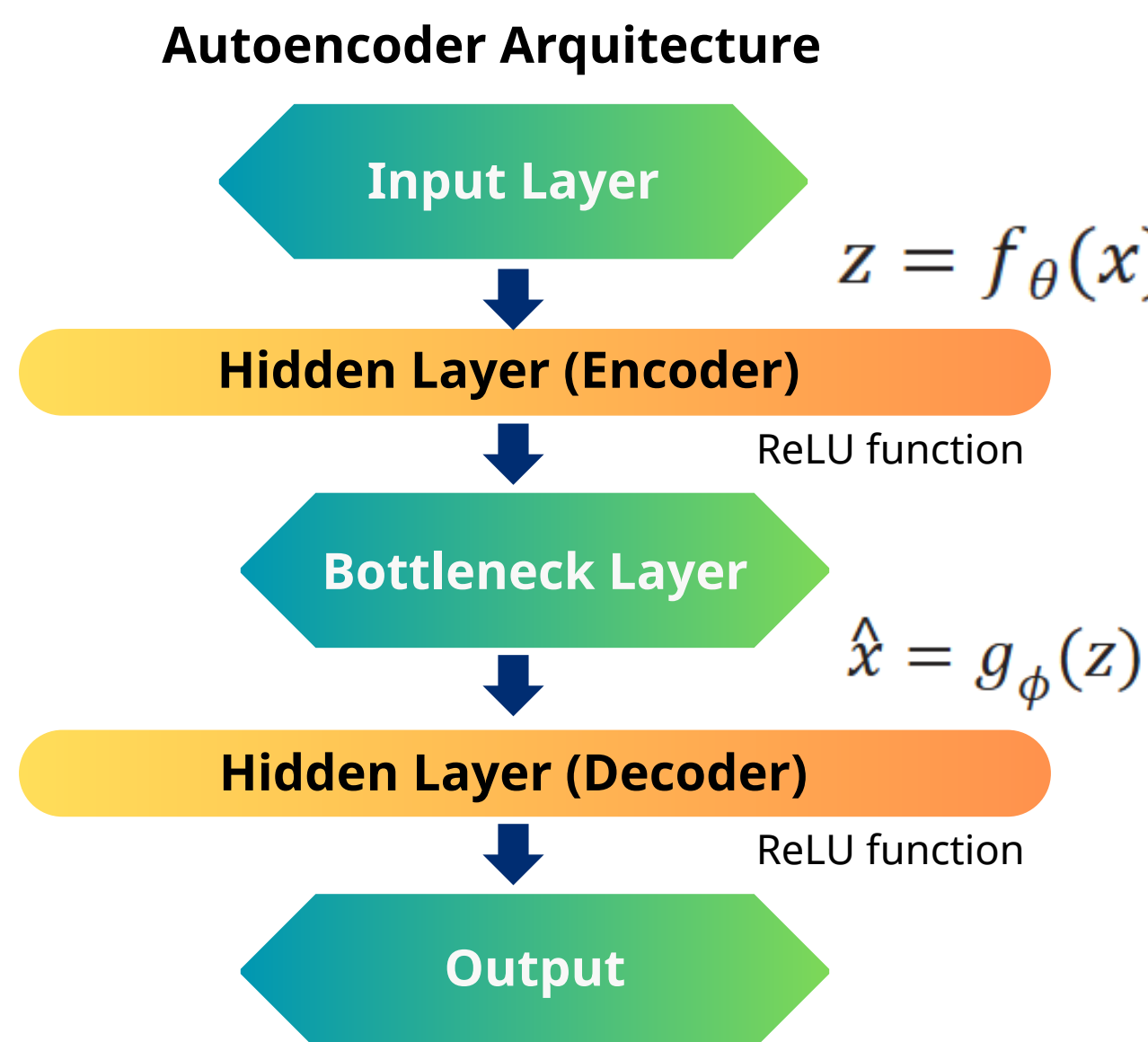
- Population density
- % population aged 65+



- Food affordability is multi-dimensional (cost, access, mobility)
- Evidence of non-linear patterns and skewed distributions
- Regional disparities suggest potential spatial dependence

PCA and Autoencoders

We compared PCA (a linear method) with autoencoders (a non-linear method) in terms of reconstruction error across different latent dimensions (2D, 3D, 4D). Lower error indicates a better ability to capture the structure of the data.



The superior performance of autoencoders suggests that the relationships among affordability variables are non-linear and cannot be fully captured by linear projections such as PCA.

Predictive Modeling

Experimental Design:

Benchmarked three feature representations using XGBoost with State-Blocked Nested Cross-Validation to ensure geographic generalizability:

- Exp 1: Socioeconomic Controls only.
- Exp 2: Raw Food Metrics + Controls + Spatial Lags
- Exp 3: 3D Embeddings + Controls + Spatial Lags

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad Y = \beta_0 + F(C, S, A) + \epsilon \quad F(C, S, A) = \sum_{k=1}^K \eta \cdot f_k(C, S, A)$$

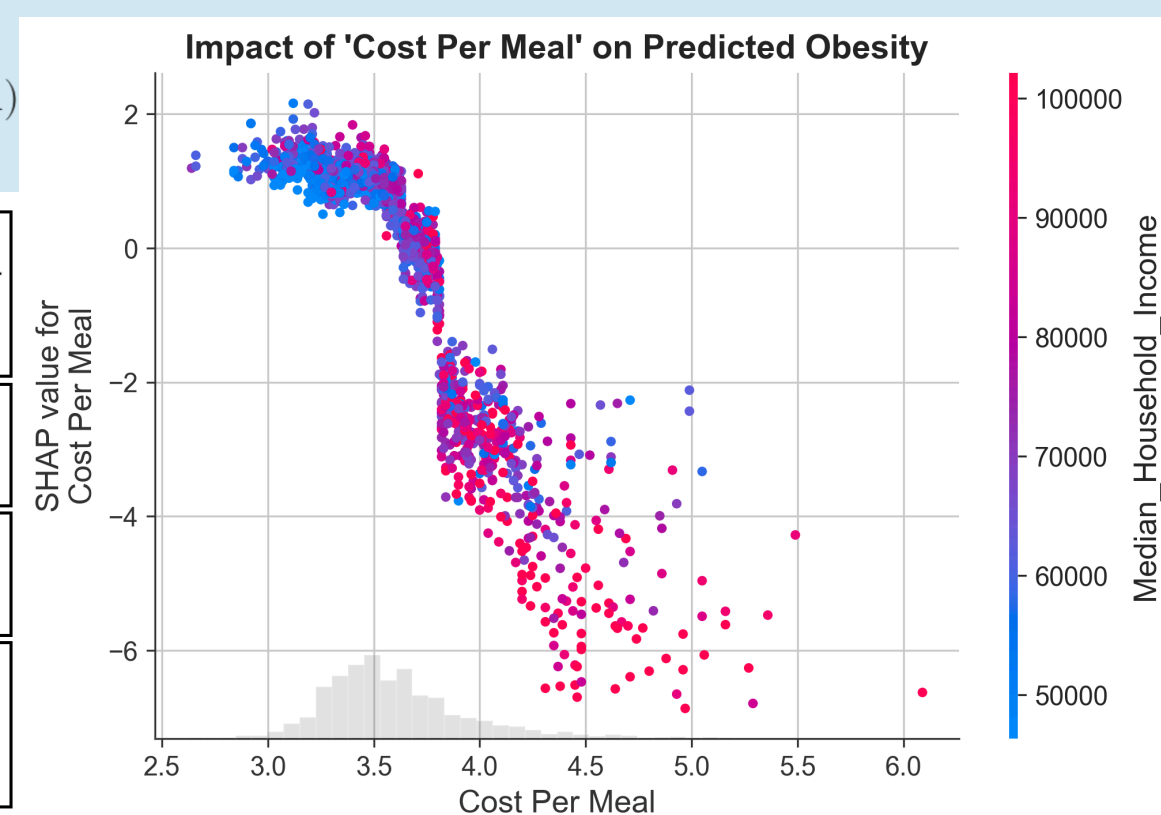
Health Outcome	Exp 1: Controls Only	Exp 2: Raw + Spatial + Controls	Exp 3: AE-3D + Spatial + Controls
Obesity Rate	0.4487	0.5877	0.5912
Diabetes Rate	0.6747	0.7260	0.7301
Medicare Cost/Capita	0.1003	0.2912	0.2985

Policy Tipping Point:

SHAP analysis identifies a critical threshold at around \$3.50 per meal.

Below this price, lower food costs are associated with an increase in obesity risk.

Once costs exceed, the factor becomes "protective," correlating with a reduction in predicted obesity rates.



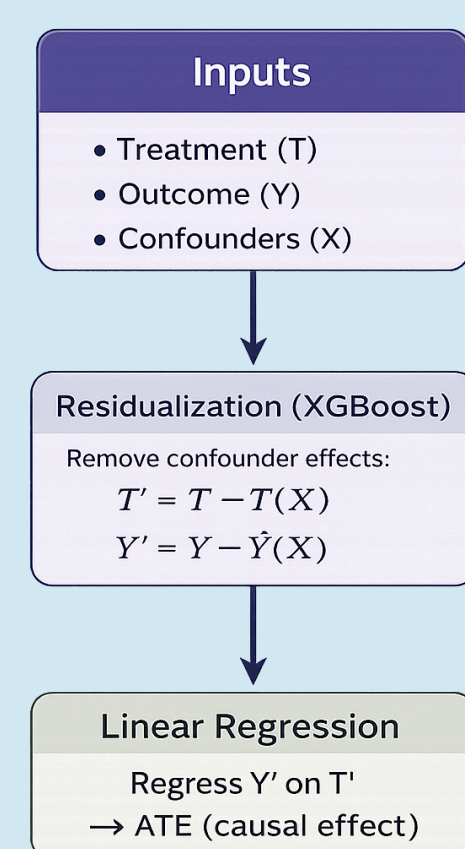
Causal Inference with DML

The Goal: Transition from predictive correlation to isolating the causal effect of specific, policy-changeable levers.

Double Machine Learning Methodology:

- Framework: Used EconML to implement a residual-on-residual causal model.
- Bias Reduction: XGBoost "nuisance" models orthogonalize treatments and outcomes, isolating true causal effects from high-dimensional noise.

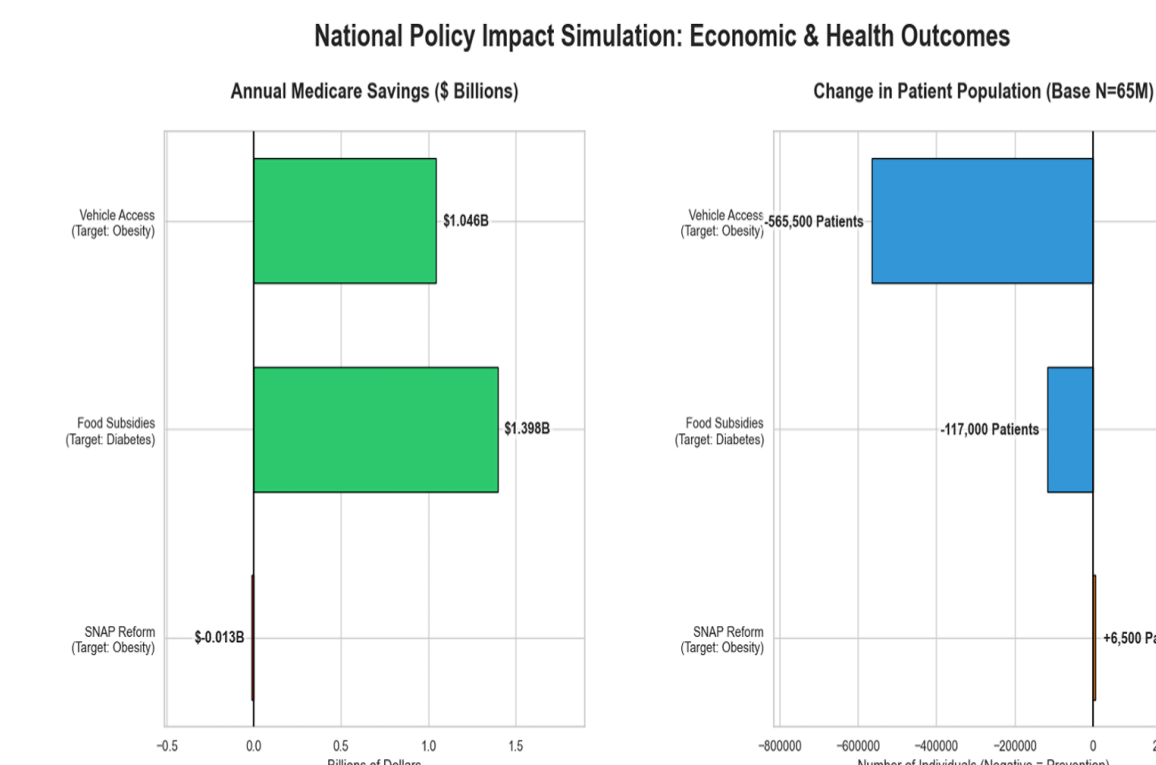
Treatment Lever	Target Outcome	Average Treatment Effect (ATE)	95% Confidence Interval
Lack of Vehicle Access	Obesity Rate (%)	+8.660	[2.580, 14.740]
Lack of Vehicle Access	Diabetes Rate (%)	+2.248	[0.028, 4.469]
Cost Per Meal	Diabetes Rate (%)	-1.795	[-2.886, -0.704]
SNAP Participation Rate	Obesity Rate (%)	+0.119	[0.074, 0.164]
SNAP Participation Rate	Diabetes Rate (%)	+0.041	[0.027, 0.055]



Conclusions

Simulation Methodology: To quantify the fiscal return on intervention, a standardized 10% policy shift was simulated. The results were calculated by mapping the DML-derived causal effects onto a base Medicare population of 65 Million individuals.

$$\Delta \text{Patients} = \text{ATE} \times \Delta \text{Policy} \times \text{Population Base}$$
$$\text{Savings} = (\Delta \text{Patients}) \times (\text{Average Annual Care Cost per Patient})$$



- Food affordability is multi-dimensional: access matters more than price alone

- Transportation access is a key factor influencing obesity and diabetes outcomes.

Acknowledgement

- Thanks to Anthony Kearsley and Dilshad Raihan, and the Johns Hopkins Department of Applied Mathematics and Statistics.
- We acknowledge the USDA, CDC, CMS, ACS for their public data and tools.